DARSHAN UNIVERSITY

Department of Computer Science & Engineering

Course Code: 2601CS102

Web Designing

B.Tech. First Semester

CHAPTER 2

Bootstrap — Responsive Front-End Framework

E-Notes / Study Material

Topic	Coverage
Introduction & Setup	What is Bootstrap, download/CDN methods
Typography, Images, Figures	Text styles, responsive images, figure captions
Tables & Container	Table styles, container / container-fluid
Breakpoints & Grid System	Responsive breakpoints, 12-column grid
Forms	Form controls, layouts, validation styles
Components	Navbar, cards, alerts, modals, carousel & more
Helper Classes & Utilities	Spacing, colors, display, flex utilities
Icon Libraries	Font Awesome installation & usage

Academic Year 2025–26

1. Introduction to Bootstrap

1.1 What is Bootstrap?

Bootstrap is a free, open-source front-end framework used for building fast, responsive, and mobile-first websites. It was originally developed by Twitter engineers Mark Otto and Jacob Thornton and released in August 2011 — which is why it's sometimes called 'Twitter Bootstrap'. Instead of writing every CSS rule from scratch, developers use Bootstrap's ready-made classes to quickly build professional-looking layouts.

Why Use Bootstrap?

Responsive by Default — Websites automatically adapt to mobile, tablet, and desktop screens
Faster Development — Pre-built components save huge amounts of coding time
Consistency — Uniform design language across all pages of a website
Cross-Browser Compatible — Works reliably in Chrome, Firefox, Safari, and Edge
Huge Community — Extensive documentation, tutorials, and third-party themes
Customizable — Can override default styles with your own custom CSS

1.2 Bootstrap Versions

Version	Release Year	Key Highlight
Bootstrap 2	2012	Introduced responsive design with a 12-column grid
Bootstrap 3	2013	Mobile-first approach, flat design
Bootstrap 4	2018	Switched from Less to Sass, added Flexbox
Bootstrap 5	2021	Removed jQuery dependency, added CSS Grid support, new utility API

Note for This Course:

We will be using Bootstrap 5 — the current, modern version.
Unlike Bootstrap 4, Bootstrap 5 does NOT require jQuery for its components to work.

2. Download and Load Bootstrap

There are two main ways to add Bootstrap to your HTML project: using a CDN link (recommended for beginners) or downloading the files locally.

2.1 Method 1 — Using CDN (Recommended)

A CDN (Content Delivery Network) hosts the Bootstrap files on fast global servers. You simply link to them — no downloading or setup required. This is the fastest and easiest way to get started.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Bootstrap Page</title>

  <!-- Bootstrap 5 CSS (place in <head>) -->
  <link
    href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
    rel="stylesheet">
</head>
<body>

  <h1 class="text-primary text-center mt-5">Hello, Bootstrap!</h1>

  <!-- Bootstrap 5 JS Bundle (place before closing </body> tag) -->
  <script

src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
>
  </script>
</body>
</html>
```

> ⚠ **Important — Two Separate Files:**
> CSS file → goes inside <head>, controls all Bootstrap styling
> JS Bundle file → goes just before </body>, powers interactive components (modals, dropdowns, carousels)
> The JS Bundle already includes Popper.js — no separate download needed

2.2 Method 2 — Download Locally

For offline projects or more control over versioning, you can download Bootstrap and host the files yourself:

1. Visit the official site: getbootstrap.com
2. Click Download → Download Compiled CSS and JS
3. Extract the ZIP file — you'll get a css/ and js/ folder
4. Copy these folders into your project directory
5. Link them using relative paths instead of CDN URLs

```
<!-- Local file linking -->
<link href="css/bootstrap.min.css" rel="stylesheet">
<script src="js/bootstrap.bundle.min.js"></script>
```

2.3 CDN vs Local — Which to Choose?

Factor	CDN Method	Local Download
Setup Speed	Instant — just copy a link	Requires downloading & folder setup
Internet Required?	Yes, to load the files	No — works fully offline
Caching Benefit	Browser may cache from other sites	No shared cache benefit
Best For	Learning, quick prototypes, most websites	Offline apps, strict environments

3. Typography, Images & Figures

3.1 Bootstrap Typography

Bootstrap provides a set of ready-made classes to style headings, paragraphs, and text elements consistently across your site.

```
<h1>h1. Bootstrap Heading</h1>
<h2>h2. Bootstrap Heading</h2>
<h3>h3. Bootstrap Heading</h3>

<!-- Display headings — larger & bolder than normal headings -->
<h1 class="display-1">Display 1</h1>
<h1 class="display-4">Display 4</h1>

<!-- Lead paragraph — larger, stands out from body text -->
<p class="lead">This is a lead paragraph. It stands out from other text.</p>

<!-- Text alignment -->
<p class="text-start">Left aligned text</p>
<p class="text-center">Center aligned text</p>
<p class="text-end">Right aligned text</p>

<!-- Text styling -->
<p><strong>Bold text</strong> and <em>italic text</em></p>
<p class="text-muted">Muted (grayed out) text</p>
<p class="text-uppercase">uppercase text</p>
<p class="text-lowercase">LOWERCASE TEXT</p>
<p class="text-capitalize">capitalize each word</p>

<!-- Blockquote -->
<blockquote class="blockquote">
  <p>A well-known quote goes here.</p>
  <footer class="blockquote-footer">Someone famous</footer>
</blockquote>
```

Class	Purpose
.lead	Makes a paragraph stand out with larger font
.display-1 to .display-6	Extra-large, bold display headings
.text-muted	Light gray secondary text
.text-start / .text-center / .text-end	Text alignment (replaces old .text-left/.text-right)
.fw-bold / .fw-normal / .fw-light	Font weight control
.fst-italic	Italic font style
.blockquote	Styled quotation block

3.2 Responsive Images

Bootstrap makes images automatically responsive (scale with their parent container) using a single utility class:

```
<!-- Responsive image — scales with parent, never overflows -->
<img src="campus.jpg" class="img-fluid" alt="University Campus">

<!-- Rounded corners -->
<img src="photo.jpg" class="img-fluid rounded" alt="Rounded image">

<!-- Perfectly circular image (great for profile pictures) -->
<img src="avatar.jpg" class="img-fluid rounded-circle" alt="Profile picture">

<!-- Thumbnail style — adds padding + light border -->
<img src="thumb.jpg" class="img-thumbnail" alt="Thumbnail">

<!-- Image alignment -->
<img src="logo.png" class="img-fluid float-start me-3" alt="Left aligned">
<img src="logo.png" class="img-fluid mx-auto d-block" alt="Center aligned">
```

Golden Rule:

Always add class="img-fluid" to every image in a Bootstrap project.

It sets max-width: 100%; height: auto; — preventing images from breaking your layout on small screens.

3.3 Figures

The <figure> component pairs an image with a caption — perfect for photo galleries, diagrams, or illustrations that need explanatory text.

```
<figure class="figure">
  <img src="chart.jpg" class="figure-img img-fluid rounded" alt="Sales Chart">
  <figcaption class="figure-caption text-center">
    Fig 1.1 — Quarterly Sales Growth Chart
  </figcaption>
</figure>
```

4. Tables & Containers

4.1 Bootstrap Tables

Bootstrap transforms plain HTML tables into clean, professional-looking data grids using simple utility classes.

```
<table class="table">
  <thead>
    <tr>
      <th scope="col">#</th>
      <th scope="col">Name</th>
      <th scope="col">Branch</th>
      <th scope="col">Grade</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <th scope="row">1</th>
      <td>Rahul Patel</td>
      <td>CSE</td>
      <td>A+</td>
    </tr>
    <tr>
      <th scope="row">2</th>
      <td>Priya Shah</td>
      <td>IT</td>
      <td>A</td>
    </tr>
  </tbody>
</table>
```

Class	Effect
.table	Base Bootstrap table styling — clean spacing & borders
.table-striped	Adds zebra-striping to alternating rows
.table-bordered	Adds borders on all sides of every cell
.table-borderless	Removes all borders completely
.table-hover	Highlights a row when the mouse hovers over it
.table-dark	Applies a dark background theme to the table
.table-sm	Reduces cell padding for a more compact table
.table-responsive	Wraps table in a scrollable container on small screens

Contextual row/cell coloring classes (great for status indicators):

```
<table class="table">
  <tbody>
    <tr class="table-success"><td>Passed</td><td>Excellent Score</td></tr>
    <tr class="table-warning"><td>Pending</td><td>Awaiting Review</td></tr>
    <tr class="table-danger"><td>Failed</td><td>Needs Improvement</td></tr>
  </tbody>
</table>
```

```
<!-- Wrap in this div for horizontal scroll on mobile -->
<div class="table-responsive">
  <table class="table">...</table>
</div>
```

4.2 Containers

A container is the fundamental building block of Bootstrap layouts. It wraps your content, adds consistent left/right padding, and centers it on the page. Every grid system in Bootstrap MUST be placed inside a container.

Container Class	Behavior
.container	Fixed max-width at each breakpoint — has visible left/right margins
.container-fluid	Always 100% width — spans the full browser window at every size
.container-sm/md/lg/xl/xxl	Fixed width starting from that specific breakpoint upward, fluid (100%) below it

```
<!-- Fixed-width, centered container (most common choice) -->
<div class="container">
  <h1>My Content</h1>
</div>

<!-- Always full width — edge to edge -->
<div class="container-fluid">
  <h1>Full Width Content</h1>
</div>

<!-- Fluid until "md" breakpoint, then fixed-width -->
<div class="container-md">
  <h1>Responsive Container</h1>
</div>
```

Rule of Thumb:

Use .container for most normal page content (blogs, forms, cards).

Use .container-fluid for full-bleed sections like hero banners or navbars that need edge-to-edge color.

5. Breakpoints & The Grid System

5.1 Responsive Breakpoints

A breakpoint is a specific screen width at which your layout changes to fit the device better. Bootstrap 5 defines six breakpoints, each identified by a short class infix:

Breakpoint	Class Infix	Screen Width	Typical Device
Extra small	(none)	< 576px	Small phones
Small	sm	≥ 576px	Large phones
Medium	md	≥ 768px	Tablets
Large	lg	≥ 992px	Laptops
Extra large	xl	≥ 1200px	Desktop monitors
Extra extra large	xxl	≥ 1400px	Large desktop monitors

How Breakpoints Work in Classes:

Classes follow the pattern: {property}-{breakpoint}-{value}
Example: col-md-6 → "On medium screens and up, take 6 of 12 grid columns"
No infix = applies to ALL screen sizes (mobile-first base rule)

5.2 The 12-Column Grid System

Bootstrap's layout engine divides every row into 12 equal-width columns. You decide how many of those 12 columns each element should occupy — this is what makes responsive design so easy.

Grid Structure — 3 Required Levels:

.container → the outer wrapper that centers and pads your content
.row → a horizontal group that holds columns (uses flexbox internally)
.col / .col-* → the actual content columns, must total 12 or less per row

```
<div class="container">

  <!-- Equal-width columns — auto divides space equally -->
  <div class="row">
    <div class="col">Column 1</div>
    <div class="col">Column 2</div>
    <div class="col">Column 3</div>
  </div>

  <!-- Specific column widths (must add up to 12) -->
  <div class="row">
    <div class="col-4">Takes 4/12 width</div>
    <div class="col-8">Takes 8/12 width</div>
  </div>
```

```
  <!-- Responsive columns — different width per breakpoint -->
  <div class="row">
    <div class="col-12 col-md-6 col-lg-4">
      Full width on mobile, half on tablet, one-third on desktop
    </div>
    <div class="col-12 col-md-6 col-lg-4">
      Same responsive behavior
    </div>
    <div class="col-12 col-md-12 col-lg-4">
      Full width on tablet, one-third on desktop
    </div>
  </div>

</div>
```

▸ Grid Gutters & Alignment

```
<!-- Remove gutters (spacing) between columns -->
<div class="row g-0">...</div>

<!-- Custom gutter size (0 to 5) -->
<div class="row g-3">...</div>

<!-- Vertically center columns -->
<div class="row align-items-center">...</div>

<!-- Horizontally distribute columns -->
<div class="row justify-content-between">...</div>
```

Common Grid Pattern	Column Classes
Two equal halves	col-md-6 + col-md-6
Sidebar + Main content	col-md-3 + col-md-9
Three equal cards	col-md-4 + col-md-4 + col-md-4
Four equal cards	col-md-3 × 4
Mobile stack, desktop row	col-12 col-md-6

6. Designing Forms with Bootstrap

Bootstrap provides ready-made classes to make form elements consistent, accessible, and attractive across your entire website.

6.1 Basic Form Controls

```
<form>
  <!-- Text input with label -->
  <div class="mb-3">
    <label for="fullName" class="form-label">Full Name</label>
    <input type="text" class="form-control" id="fullName" placeholder="Enter your
name">
  </div>

  <!-- Email input -->
  <div class="mb-3">
    <label for="email" class="form-label">Email Address</label>
    <input type="email" class="form-control" id="email"
placeholder="name@example.com">
  </div>

  <!-- Select dropdown -->
  <div class="mb-3">
    <label for="branch" class="form-label">Branch</label>
    <select class="form-select" id="branch">
      <option selected>Choose Branch...</option>
      <option value="cse">Computer Science</option>
      <option value="it">Information Technology</option>
    </select>
  </div>

  <!-- Textarea -->
  <div class="mb-3">
    <label for="message" class="form-label">Message</label>
    <textarea class="form-control" id="message" rows="3"></textarea>
  </div>

  <!-- Checkbox -->
  <div class="form-check mb-3">
    <input class="form-check-input" type="checkbox" id="agree">
    <label class="form-check-label" for="agree">I agree to the terms</label>
  </div>

  <!-- Radio buttons -->
  <div class="form-check">
    <input class="form-check-input" type="radio" name="gender" id="male">
    <label class="form-check-label" for="male">Male</label>
  </div>
  <div class="form-check mb-3">
    <input class="form-check-input" type="radio" name="gender" id="female">
    <label class="form-check-label" for="female">Female</label>
  </div>

  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

Class	Applies To
.form-control	Text inputs, textareas, password, email, number fields
.form-select	Dropdown <select> elements
.form-check + .form-check-input	Checkboxes and radio buttons
.form-label	Labels for form fields
.form-text	Small helper/hint text below a field
.form-switch	Turns a checkbox into a toggle switch style

6.2 Form Layout Options

```
<!-- Horizontal form layout using the grid -->
<form>
  <div class="row mb-3">
    <label for="name2" class="col-sm-2 col-form-label">Name</label>
    <div class="col-sm-10">
      <input type="text" class="form-control" id="name2">
    </div>
  </div>
</form>

<!-- Inline form elements -->
<form class="row row-cols-lg-auto g-3 align-items-center">
  <div class="col-12">
    <input type="text" class="form-control" placeholder="Search...">
  </div>
  <div class="col-12">
    <button type="submit" class="btn btn-primary">Go</button>
  </div>
</form>
```

6.3 Form Validation Styles

Bootstrap provides built-in visual feedback classes for form validation (used together with HTML5 or JavaScript validation logic):

```
<div class="mb-3">
  <label for="validEmail" class="form-label">Email</label>
  <input type="email" class="form-control is-valid" id="validEmail"
value="test@mail.com">
  <div class="valid-feedback">Looks good!</div>
</div>

<div class="mb-3">
  <label for="invalidEmail" class="form-label">Email</label>
  <input type="email" class="form-control is-invalid" id="invalidEmail">
  <div class="invalid-feedback">Please enter a valid email.</div>
</div>
```

Key Validation Classes:

.is-valid / .valid-feedback → green border + success message (shown when valid)

.is-invalid / .invalid-feedback → red border + error message (shown when invalid)

7. Components of Bootstrap

Bootstrap ships with dozens of pre-built, reusable UI components. Below are the most commonly used ones in real projects.

7.1 Navbar

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand" href="#">MySite</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
            data-bs-target="#navMenu">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navMenu">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#">About</a></li>
        <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
      </ul>
    </div>
  </div>
</nav>
```

7.2 Cards

```
<div class="card" style="width: 18rem;">
  <img src="movie.jpg" class="card-img-top" alt="Movie Poster">
  <div class="card-body">
    <h5 class="card-title">Inception</h5>
    <p class="card-text">A thief who steals corporate secrets through dream-
sharing technology.</p>
    <a href="#" class="btn btn-primary">View Details</a>
  </div>
</div>
```

7.3 Buttons & Button Groups

```
<button class="btn btn-primary">Primary</button>
<button class="btn btn-success">Success</button>
<button class="btn btn-danger">Danger</button>
<button class="btn btn-outline-secondary">Outline</button>

<!-- Sizes -->
<button class="btn btn-primary btn-lg">Large Button</button>
<button class="btn btn-primary btn-sm">Small Button</button>

<!-- Button group -->
<div class="btn-group" role="group">
  <button class="btn btn-outline-primary">Left</button>
  <button class="btn btn-outline-primary">Middle</button>
  <button class="btn btn-outline-primary">Right</button>
</div>
```

7.4 Alerts

```
<div class="alert alert-success" role="alert">
  Your form was submitted successfully!
</div>
<div class="alert alert-danger alert-dismissible fade show" role="alert">
  Error! Please check your input.
  <button type="button" class="btn-close" data-bs-dismiss="alert"></button>
</div>
```

7.5 Modal (Popup Dialog)

```
<!-- Trigger button -->
<button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#myModal">
  Open Modal
</button>

<!-- Modal structure -->
<div class="modal fade" id="myModal" tabindex="-1">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Movie Details</h5>
        <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body">
        Full movie synopsis and cast information goes here.
      </div>
      <div class="modal-footer">
        <button class="btn btn-secondary" data-bs-dismiss="modal">Close</button>
      </div>
    </div>
  </div>
</div>
```

7.6 Carousel (Image Slider)

```
<div id="myCarousel" class="carousel slide" data-bs-ride="carousel">
  <div class="carousel-inner">
    <div class="carousel-item active">
      <img src="slide1.jpg" class="d-block w-100" alt="Slide 1">
    </div>
    <div class="carousel-item">
      <img src="slide2.jpg" class="d-block w-100" alt="Slide 2">
    </div>
  </div>
  <button class="carousel-control-prev" type="button" data-bs-target="#myCarousel"
data-bs-slide="prev">
    <span class="carousel-control-prev-icon"></span>
  </button>
  <button class="carousel-control-next" type="button" data-bs-target="#myCarousel"
data-bs-slide="next">
    <span class="carousel-control-next-icon"></span>
  </button>
</div>
```

7.7 Other Essential Components — Quick Reference

Component	Purpose
Badge	Small count/label indicator, e.g. notification counts
Breadcrumb	Shows navigation path: Home > Courses > Web Designing
Accordion	Collapsible content panels (FAQ sections)
Dropdown	Toggleable menu of links or actions
Pagination	Page number navigation links
Progress Bar	Visual indicator of task/loading completion
Spinner	Loading animation indicator
Tooltip / Popover	Small info box shown on hover/click
Toast	Small non-blocking notification message
List Group	Styled list of items, often used for menus

8. Helper Classes & Utilities in Bootstrap

Bootstrap includes hundreds of small single-purpose utility classes. These let you style elements directly in HTML without writing custom CSS — a technique known as 'utility-first' styling.

8.1 Spacing Utilities (Margin & Padding)

Spacing classes follow the pattern: {property}{sides}-{breakpoint}-{size}

Part	Options	Meaning
Property	m (margin) | p (padding)	Which spacing property to apply
Sides	t, b, s, e, x, y, (blank=all)	top, bottom, start(left), end(right), x-axis, y-axis, all sides
Size	0, 1, 2, 3, 4, 5, auto	0=none → 5=largest (based on rem units)

```
<div class="m-3">Margin on all sides</div>
<div class="mt-4">Margin top only</div>
<div class="px-2">Padding left & right (x-axis)</div>
<div class="py-5">Padding top & bottom (y-axis)</div>
<div class="mb-0">Zero margin bottom</div>
<div class="mx-auto" style="width: 200px;">Horizontally centered block</div>
```

8.2 Color Utilities

```
<!-- Text colors -->
<p class="text-primary">Primary blue text</p>
<p class="text-success">Success green text</p>
<p class="text-danger">Danger red text</p>
<p class="text-warning">Warning yellow text</p>

<!-- Background colors -->
<div class="bg-primary text-white p-3">Primary background</div>
<div class="bg-light p-3">Light gray background</div>
<div class="bg-dark text-white p-3">Dark background</div>
```

Theme Color Name	Typical Use Case
primary	Main brand color, primary buttons/links
secondary	Secondary/muted actions
success	Positive actions, confirmations
danger	Errors, destructive actions
warning	Cautions, alerts
info	Informational messages
light / dark	Light or dark backgrounds

8.3 Display & Flexbox Utilities

```
<!-- Display -->
<div class="d-none">Hidden</div>
<div class="d-block">Block-level</div>
<div class="d-none d-md-block">Hidden on mobile, visible from md upward</div>

<!-- Flexbox -->
<div class="d-flex justify-content-between align-items-center">
  <span>Left Item</span>
  <span>Right Item</span>
</div>

<div class="d-flex flex-column">
  <div>Stacked Item 1</div>
  <div>Stacked Item 2</div>
</div>
```

Utility Class	Effect
d-flex	Turns element into a flex container
justify-content-center / between / around	Horizontal alignment of flex items
align-items-center / start / end	Vertical alignment of flex items
flex-row / flex-column	Direction of flex items
flex-wrap	Allows flex items to wrap onto new lines

8.4 Other Frequently Used Utilities

Class	Purpose
.text-center / .text-start / .text-end	Text alignment
.w-25 / .w-50 / .w-75 / .w-100	Set width as % of parent (25%, 50%, 75%, 100%)
.h-100	Set height to 100% of parent
.rounded / .rounded-circle / .rounded-0	Border radius control
.shadow / .shadow-sm / .shadow-lg	Box shadow depth
.border / .border-primary / .border-0	Add or remove borders
.position-relative / .position-absolute	CSS position control
.overflow-hidden / .overflow-auto	Control content overflow behavior
.visually-hidden	Hides element visually but keeps it for screen readers

9. Loading Icon Libraries — Font Awesome

Icons make interfaces more intuitive and visually appealing. Font Awesome is the most popular icon library on the web, offering thousands of scalable vector icons that work perfectly alongside Bootstrap.

9.1 Adding Font Awesome via CDN

Just like Bootstrap, the easiest way to use Font Awesome is through a CDN link placed in your <head> section:

```
<head>
  <!-- Bootstrap CSS -->
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
        rel="stylesheet">

  <!-- Font Awesome CDN -->
  <link rel="stylesheet"
        href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.5.1/css/all.min.css">
</head>
```

9.2 Using Font Awesome Icons

Once loaded, insert any icon using a simple <i> tag with the correct class name:

```
<!-- Basic icon usage -->
<i class="fa-solid fa-house"></i>
<i class="fa-solid fa-user"></i>
<i class="fa-solid fa-envelope"></i>
<i class="fa-brands fa-instagram"></i>

<!-- Icon inside a Bootstrap button -->
<button class="btn btn-primary">
  <i class="fa-solid fa-download"></i> Download
</button>

<!-- Icon sizing -->
<i class="fa-solid fa-star fa-lg"></i>   <!-- Large -->
<i class="fa-solid fa-star fa-2x"></i>   <!-- 2x size -->
<i class="fa-solid fa-star fa-3x"></i>   <!-- 3x size -->

<!-- Icon coloring using Bootstrap text color classes -->
<i class="fa-solid fa-heart text-danger fa-2x"></i>

<!-- Spinning loading icon -->
<i class="fa-solid fa-spinner fa-spin"></i>
```

Style Prefix	Icon Category	Example
fa-solid	Solid, filled icons (most common)	fa-solid fa-house
fa-regular	Outlined, lighter-weight icons	fa-regular fa-heart

fa-brands	Brand/logo icons (social media, companies)	fa-brands fa-github

How to Find Icon Names:

Visit fontawesome.com/icons — search for any icon you need.

Click the icon to copy its exact class name (e.g., fa-solid fa-magnifying-glass).

Paste it directly into an <i> or tag in your HTML.

9.3 Font Awesome + Bootstrap Navbar Example

```
<nav class="navbar navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand" href="#">
      <i class="fa-solid fa-film"></i> MovieHub
    </a>
    <ul class="navbar-nav flex-row gap-3">
      <li><a class="nav-link" href="#"><i class="fa-solid fa-house"></i></a></li>
      <li><a class="nav-link" href="#"><i class="fa-solid fa-magnifying-
glass"></i></a></li>
      <li><a class="nav-link" href="#"><i class="fa-solid fa-heart"></i></a></li>
      <li><a class="nav-link" href="#"><i class="fa-solid fa-user"></i></a></li>
    </ul>
  </div>
</nav>
```

Quick Reference — Bootstrap Cheat Sheet

Setup

Task	Code Snippet
Load CSS	<link href="...bootstrap.min.css" rel="stylesheet">
Load JS Bundle	<script src="...bootstrap.bundle.min.js"></script>

Grid

Class	Purpose
.container / .container-fluid	Fixed-width or full-width wrapper
.row	Horizontal group of columns
.col-*, .col-md-*, .col-lg-*	Responsive column widths (out of 12)

Typography & Images

Class	Purpose
.lead	Larger intro paragraph
.text-muted	Gray secondary text
.img-fluid	Responsive image scaling
.rounded-circle	Circular image

Tables

Class	Purpose
.table	Base table style
.table-striped	Zebra-striped rows
.table-hover	Row highlight on hover
.table-responsive	Horizontal scroll on small screens

Forms

Class	Purpose
.form-control	Text/email/password input styling
.form-select	Dropdown select styling
.form-check-input	Checkbox/radio styling
.is-valid / .is-invalid	Validation feedback styling

Components

Component	Key Classes
Navbar	.navbar .navbar-expand-lg .navbar-brand
Card	.card .card-body .card-title .card-text
Button	.btn .btn-primary .btn-outline-*
Alert	.alert .alert-success/danger/warning
Modal	.modal .modal-dialog .modal-content
Carousel	.carousel .carousel-item .carousel-inner

Spacing & Utilities

Class Pattern	Purpose
m{side}-{size} / p{side}-{size}	Margin / padding (0–5, auto)
.d-flex / .justify-content-* / .align-items-*	Flexbox layout control
.text-center/start/end	Text alignment
.bg-primary/success/danger	Background color
.shadow / .rounded / .border	Visual embellishments

Font Awesome Icons

Snippet	Purpose
<link rel="stylesheet" href="...all.min.css">	Load Font Awesome
	Insert a solid icon
	Insert a brand/logo icon
fa-lg / fa-2x / fa-3x	Icon size modifiers

Practice Questions — Bootstrap

Instructions:

Attempt all questions. Short answer questions carry 2 marks each. Long answer / coding questions carry 5 marks each.

Section A — Short Answer Questions (2 marks each)

1. What is Bootstrap? Name any two benefits of using it.
2. Differentiate between .container and .container-fluid.
3. What is the purpose of the class img-fluid?
4. List the six responsive breakpoints in Bootstrap 5, with their minimum widths.
5. What is the total number of columns in the Bootstrap grid system?
6. What is the difference between .table-striped and .table-bordered?
7. Name any four form control classes used in Bootstrap forms.
8. What is the purpose of the data-bs-toggle attribute?
9. How do you make a Bootstrap image perfectly circular?
10. What is the difference between fa-solid and fa-brands in Font Awesome?

Section B — Long Answer Questions (5 marks each)

1. Explain the Bootstrap Grid System in detail. Create a responsive 3-column layout that stacks to a single column on mobile screens. (5 marks)
2. Design a Bootstrap navigation bar with a brand logo, 4 navigation links, and a collapsible hamburger menu for mobile view. (5 marks)
3. Create a registration form using Bootstrap form classes with fields for Name, Email, Password, Gender (radio), and a Submit button. Include validation feedback styling. (5 marks)
4. Explain any five Bootstrap components with their key classes and a short code example for each. (5 marks)
5. What are helper/utility classes in Bootstrap? Explain spacing utilities and flexbox utilities with examples. (5 marks)
6. Explain the two methods of loading Bootstrap into a project. Which method would you recommend for a college project and why? (5 marks)

Section C — Coding Exercises

1. Create a product card component using Bootstrap's card class, including an image, title, description, price badge, and 'Add to Cart' button with a Font Awesome cart icon. (5 marks)
2. Build a Bootstrap table showing 5 students with columns: Roll No, Name, Branch, CGPA. Apply striped, hover, and responsive classes. (3 marks)
3. Create a Bootstrap modal that displays 'Terms & Conditions' text, triggered by a button click. (3 marks)